Travel Medicine, Part 2: Dengue, Zika, Chikungunya, Brucellosis

Dr. Majed Alnabulsi, Internal Medicine | Source: `Travel_medicine_Part_two_2.0.pdf` (37 slides)

Legend: ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes. The lecture has **no answer key** for its cases; the answers below are **[reasoning added; key not on slides]**.

🔗 **Bridge from earlier lectures:**

- **FUO & Travel Medicine Part 1 (same lecturer):**
- same "Approach to fever" slides (history, exam, DDx, work-up), repeated here as slides 27–30
- **brucellosis** was already on the FUO infection list
- malaria is still the first exclusion in a febrile traveller *[added]*
- the Part 1 Case 1 distractors (**dengue, chikungunya**) are taught here
- **Tuberculosis (Kotb): brucellar spondylitis** vs **Pott's disease** (TB spine), both chronic febrile illnesses with night sweats. **Rifampin** is shared (doxy + rifampin for brucellosis).
- **Infective Endocarditis (Reihan):** *Brucella* is a cause of **culture-negative endocarditis** *[added]* and the leading cause of brucellosis deaths.
- **Cardiogenic shock lecture:** severe dengue = **hypovolaemic shock from plasma leak** → titrated isotonic fluids, monitor haematocrit.
- **Pneumonia / ARDS:** dengue plasma leak → **pleural effusion, respiratory distress**.

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Manifestations and objective findings of travel illnesses	Dengue warning signs; Zika conjunctivitis; chikungunya arthralgia
Knowledge	⭐ Epidemiology, manifestations, complications, management	Aedes vector; congenital Zika; brucellosis exposure and regimen
Skill	Interpret manifestations and investigations	NS1/PCR early vs IgM later; rising Hct + falling platelets
Skill	Evidence-based management	No NSAIDs until dengue excluded; doxy + rifampin ≥ 6 weeks
Value	Professional attitude	—

Topics (slide 3): Dengue → Zika → Chikungunya → Brucellosis → generic "Approach to fever" (slides 27–30).

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2. Main Content (Lecture Order)

2.0 Opening Case (Slide 5 = Case Scenario 1, slide 33)

Presentation:

- **26-year-old primigravida, 22 weeks**
- vaginal bleeding + suprapubic pain for 2 h
- she has **no travel**; her **husband is a flight attendant** who often visits **South America and Southeast Asia**
- US: **fetal demise + placental detachment**
- stillborn fetus: **microcephaly**
- placental **RT-PCR: flavivirus**

Which measure in pregnancy would most likely have prevented it? → ⭐ **B. Use of condoms** *[reasoning added; key not on slides]*:

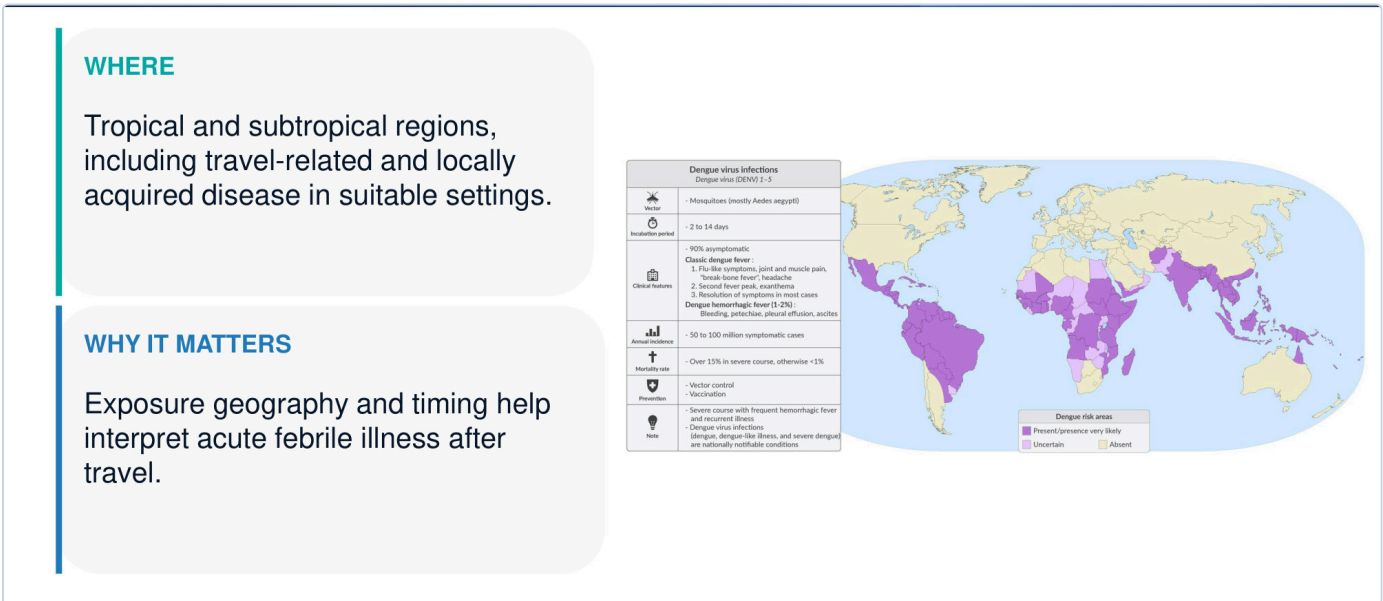
- **flavivirus + microcephaly + partner travelling to endemic areas** → **congenital Zika syndrome**
- she didn't travel, so the route is **sexual transmission** from the husband (slide 17), then **transplacental** spread
- slide 20: **use condoms or abstain** after possible exposure

Why not the others:

- **A** (unpasteurised dairy) → *Listeria/Brucella*
- **C** (ART) → HIV
- **D** (undercooked meat) → *Toxoplasma*, which can cause microcephaly but isn't a flavivirus
- **E**: **no Zika vaccine** exists

> ⚠️ Placental detachment (abruption) is **not a characteristic** Zika feature. The question hinges on the microcephaly + flavivirus.

2.1 Dengue: Epidemiology & Aetiology (Slides 6–7) ⭐



Slide 6: Dengue distribution map

Item	Detail
Where	Tropical and subtropical regions, including **travel-related** and **locally acquired** disease in suitable settings *[added: endemic in western Saudi Arabia, e.g. Jeddah, Makkah]*
Why it matters	**Exposure geography and timing** help interpret acute febrile illness after travel
Pathogen	⭐ **Dengue virus (DENV 1–4)**, **RNA virus, genus *Flavivirus***

Item	Detail
Vector	⭐ ***Aedes* mosquitoes**, particularly ***Aedes aegypti*** *[added: day-biting]*

2.2 Dengue: Classification (Slide 8) ⭐⭐



Slide 8: 2009 WHO dengue classification

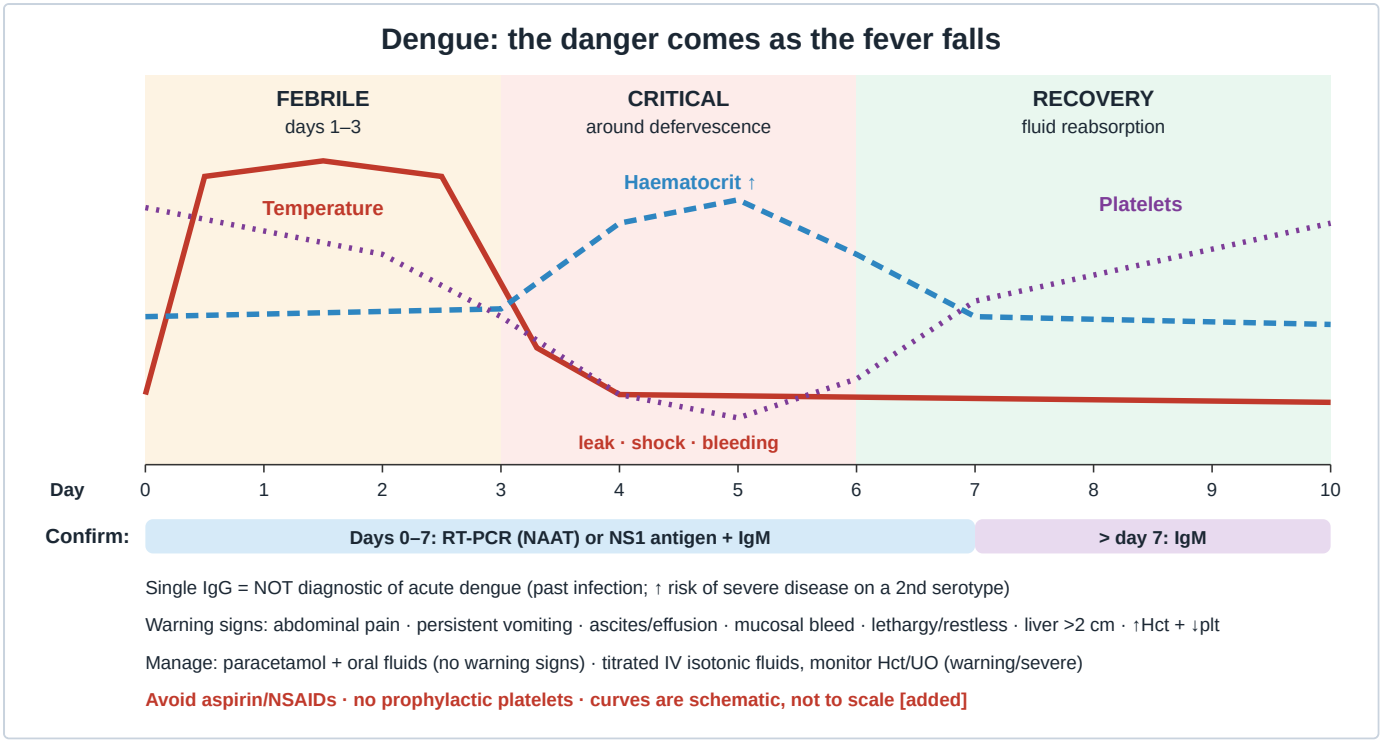
2009 WHO scheme:

Class	Slide-8 table contents
01 Without warning signs (D-W)	**Fever + 2 of:** nausea/vomiting, rash, aches and pains, **leukopenia**, **positive tourniquet test**, lab-confirmed dengue. **Ability to tolerate adequate oral fluids** and to **pass urine at least every 6 h** (i.e. outpatient-manageable)
02 With warning signs (D+W)	**≥ 1 warning sign** (abdominal pain/tenderness, persistent vomiting, clinical fluid accumulation, mucosal bleed, lethargy/restlessness, liver > 2 cm, **↑Hct with concurrent ↓platelets (≤ 100,000/mm³)**) **OR** a **comorbidity** (pregnancy, infancy, old age, diabetes, renal failure) **OR social** risk (living alone, living far from hospital)
03 Severe dengue	**Severe plasma leak → shock** or fluid accumulation with **respiratory distress**; **severe bleeding** (clinician-judged); **severe organ involvement**: liver **AST/ALT ≥ 1000 IU/L**, CNS (impaired consciousness), heart and other organs

2.3 Dengue: Clinical Phases (Slide 9) ⭐⭐

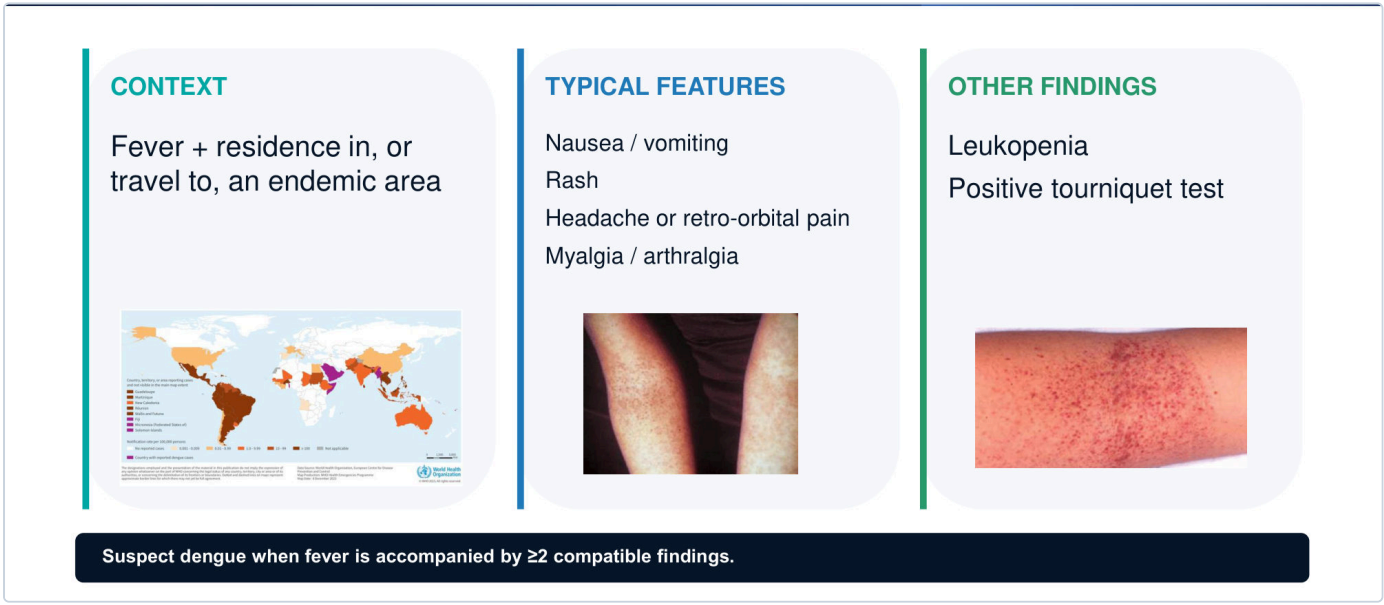
Phase	Timing	Features
Febrile	Days 1–3 (variable)	High fever, pain, rash
⭐ **Critical**	**Around defervescence**	**Plasma leakage, shock, rising haematocrit**
Recovery	Fluid reabsorption	Clinical stabilisation *[added: watch for fluid overload, bradycardia, convalescent "islands of white in a sea of red" rash]*

⭐ **Defervescence does NOT automatically mean improvement.**



Added diagram: Dengue phases, haematocrit/platelet trend and test windows

2.4 Dengue Without Warning Signs (Slide 10)



Slide 10: Dengue without warning signs: map, rash, positive tourniquet test

Context	Typical features	Other findings
Fever + **residence in or travel to an endemic area**	Nausea/vomiting; **rash**; headache or ⭐ **retro-orbital pain**; **myalgia/arthralgia** ("breakbone fever" *[added]*)	**Leukopenia**; **positive tourniquet test**

⭐ Suspect dengue when fever comes with **≥ 2 compatible findings**.

[added] **Tourniquet test:**

- inflate the BP cuff midway between SBP and DBP for **5 min**
- positive if **≥ 10–20 petechiae per square inch**

2.5 Dengue With Warning Signs (Slide 11) ⭐⭐

Domain	Warning sign
Abdominal	**Pain or tenderness**
Vomiting	**Persistent**
Leakage	**Ascites or pleural effusion**
Bleeding	**Mucosal** bleeding
Mental state	**Lethargy or restlessness**
Hepatomegaly	**Liver > 2 cm**
CBC trend	⭐ **Rising haematocrit + falling platelets**

Mnemonic, 5 B/L domains: "Leak · Bleed · Brain · Belly · Blood"

- **Leak** = ascites/effusion
- **Bleed** = mucosal
- **Brain** = lethargy/restlessness
- **Belly** = pain, persistent vomiting, liver > 2 cm
- **Blood** = ↑Hct + ↓platelets

⭐ Warning signs often appear **around defervescence**. **Reassess even when the fever improves.**

2.6 Severe Dengue (Slide 12) ⭐

Plasma leakage	Severe bleeding	Organ involvement
Shock or **respiratory distress** from fluid accumulation	Clinically significant haemorrhage	Severe hepatic, neurological, cardiac or other organ dysfunction

⭐ Severe dengue is **more likely after heterologous secondary infection** (a different serotype; *[added: antibody-dependent enhancement]*), **but can occur with primary infection**.

2.7 Dengue: Diagnostics (Slide 13) ⭐⭐

Routine labs	Confirmation
Leukopenia / neutropenia	⭐ **Days 0–7: NAAT (RT-PCR) or NS1 antigen, with IgM**
Thrombocytopenia	**After day 7: IgM** is often the primary test
Elevated AST	⭐ **A single IgG is NOT useful** for acute dengue (it reflects past infection)
Haematocrit trend (plasma leak)	

2.8 Dengue: Management (Slide 14) ⭐⭐

No warning signs	Warning / severe
Oral hydration	**Observe or admit** as appropriate
Acetaminophen (paracetamol)	**Carefully titrated IV isotonic fluids**
Close follow-up + return precautions	Monitor **vital signs, urine output, Hct**
	Blood products for significant bleeding

⭐ **Avoid aspirin/NSAIDs** (bleeding, platelet dysfunction). **No prophylactic platelet transfusions.**

[added] **Vaccines:**

- **Dengvaxia** only for seropositive people (it raises severe-dengue risk in seronegative people)
- **Qdenga (TAK-003)** is licensed in some countries
- neither is on the slides

> 🔁 **Recap: Dengue.**
> - **Organism and vector:** DENV 1–4 (flavivirus), *Aedes aegypti*.
> - **Clinical:** fever + retro-orbital pain, myalgia, rash, leukopenia, positive tourniquet.
> - **Critical phase: at defervescence**; watch **↑Hct + ↓plt**, warning signs.
> - **Testing: NS1/PCR ≤ day 7**, **IgM after**; IgG alone is useless.
> - **Treatment: paracetamol, fluids, no NSAIDs, no prophylactic platelets**.
>
> 🔗 **Link:** Zika shares the **flavivirus genus and *Aedes aegypti* vector**, which is why you "**assess for concomitant dengue before NSAIDs**" in Zika too. Zika is usually milder but **devastating in pregnancy**.
>
> ▶️ **Up next: Zika.**

2.9 Zika: Epidemiology, Aetiology, Transmission (Slides 15–17) ⭐

Item	Detail
Geography	Tropical and subtropical areas; historical outbreaks in **Africa, Southeast Asia, Pacific islands, the Americas**
History	Large outbreaks in the Americas from **2015**, notably **Brazil**; check **current travel advisories**
Pathogen	**Zika virus**, ***Flavivirus***, an **arbovirus**; **positive-sense, single-stranded, enveloped RNA**

Transmission (slide 17), three routes:

#	Route	Detail
01	**Mosquito**	***Aedes aegypti***
02	⭐ **Pregnancy**	**Transplacental → fetus**
03	⭐ **Sexual**	**Partner-to-partner**

[added] Blood transfusion is also a route.

2.10 Zika: Clinical Features (Slide 18)

Many asymptomatic	Symptoms	Rash
A **large proportion** of infections cause no symptoms *[added: ~80 %]*	**Low-grade** fever; headache, myalgia, arthralgia; ⭐ **non-purulent conjunctivitis**; malaise	Often **maculopapular and pruritic**

2.11 Zika: Complications (Slide 19) ⭐⭐

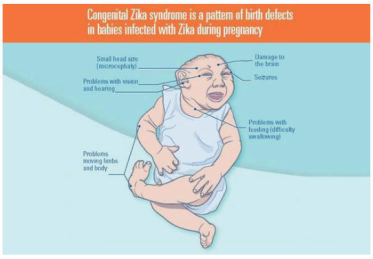
PREGNANCY

Congenital Zika syndrome

Microcephaly / CNS abnormalities

Growth restriction

Pregnancy loss



NEUROLOGIC

Guillain–Barré syndrome in infected adults

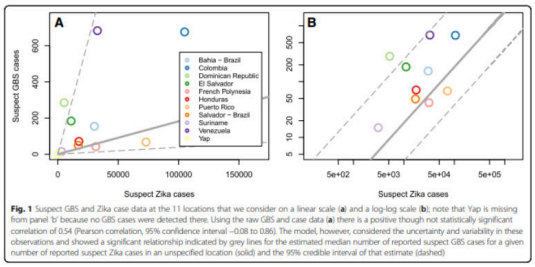


Fig. 1 Suspect GBS and Zika case data at the 11 locations that we consider on a linear scale (a) and a log-log scale (b); note that Yap is missing from panel 'b' because no GBS cases were detected there. Using the raw GBS and case data (a) there is a positive though not statistically significant correlation of 0.54 (Pearson correlation, 95% confidence interval −0.08 to 0.86). The model, however, considered the uncertainty and variability in these observations and showed a significant relationship indicated by grey lines for the estimated median number of reported suspect GBS cases for a given number of reported suspect Zika cases in an unspecified location (solid) and the 95% credible interval of that estimate (dashed)

Mier-Y-Teran-Romero L, Delorey MJ, Sejvar JJ, Johansson MA. Guillain-Barré syndrome risk among individuals infected with Zika virus: a multi-country assessment. BMC Med. 2018 May 15;16(1):67. doi: 10.1186/s12916-018-1052-4. PMID: 29759069; PMCID: PMC5952697.

Slide 19: Congenital Zika syndrome; Guillain-Barré risk data

Pregnancy: congenital Zika syndrome	Neurological (adults)
⭐ **Microcephaly / CNS abnormalities**	⭐ **Guillain–Barré syndrome**
Growth restriction	
Pregnancy loss	

The slide cites a multi-country GBS risk analysis: Mier-y-Teran-Romero et al., *BMC Med* 2018.

2.12 Zika: Management (Slide 20) ⭐

Treatment	Pregnancy & travel
No specific antiviral	⭐ **Avoid travel** to areas with active Zika notices **during pregnancy**
Supportive care	⭐ **Use condoms or abstain** after possible exposure
⭐ **Assess for concomitant dengue before using NSAIDs**	**Preconception counselling** per current guidance

Testing: consider molecular/serological tests based on **symptoms, pregnancy status and exposure**.

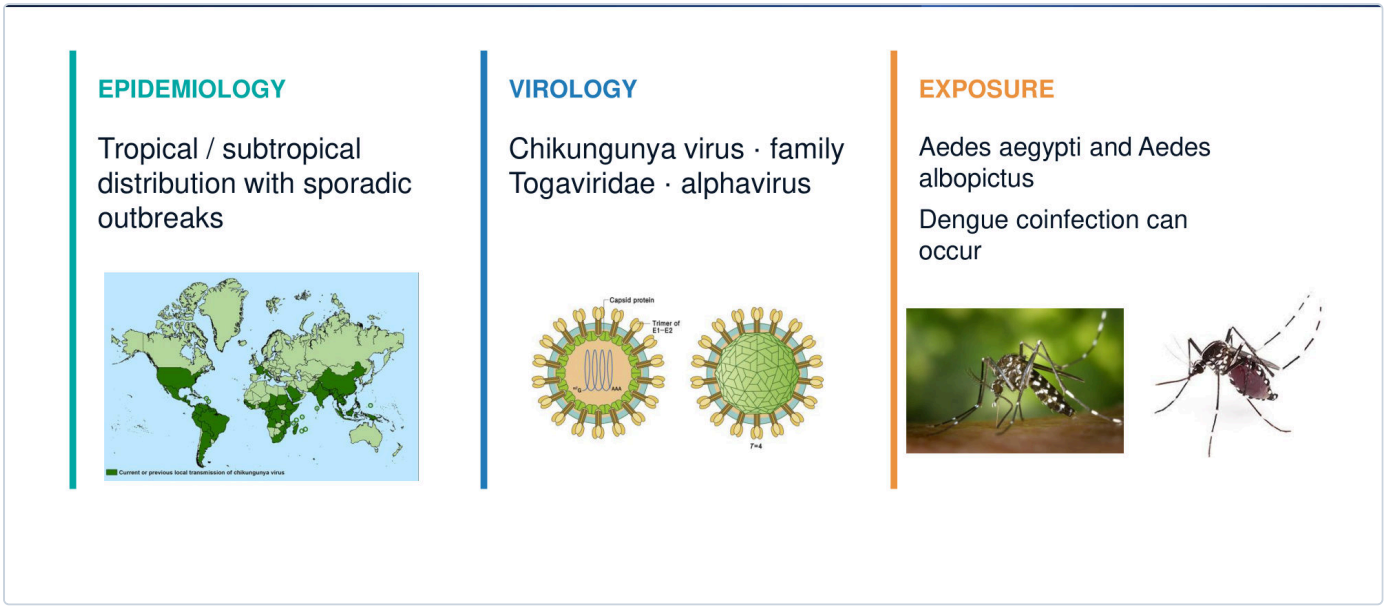
🔁 **Recap: Zika.**

- **Organism:** flavivirus; transmitted by **mosquito, transplacental and sexual** routes.
- **Clinical:** mostly asymptomatic; low fever, **conjunctivitis**, pruritic rash.
- **Complications: microcephaly**, **GBS**.
- **Prevention: condoms**, avoid travel in pregnancy.
- **Treatment:** supportive; exclude dengue before NSAIDs.

🔗 **Link:** chikungunya uses the **same *Aedes* vectors** (plus *Ae. albopictus*). It is a **different family (alphavirus)**, and its signature is **severe joint pain**.

▶️ **Up next: Chikungunya.**

2.13 Chikungunya (Slides 21–23) ⭐



Slide 21: Chikungunya virus and Aedes vectors

Item	Detail
Epidemiology	Tropical/subtropical, **sporadic outbreaks**
Virology	⭐ **Chikungunya virus**, family **Togaviridae**, **alphavirus** (not a flavivirus)
Vector	***Aedes aegypti* and *Aedes albopictus***
Coinfection	**Dengue coinfection can occur**

Classic clues	Diagnosis
Acute fever, malaise	**Exposure + clinical presentation**
⭐ **Severe polyarthralgia** *[added: symmetric, small joints of hands/feet; name = "that which bends up"]*	**RT-PCR early**
Headache, myalgia	**Serology later**
Possible lymphadenopathy	

⭐ **Joint symptoms may persist for months or longer**, even after the fever resolves.

Management:

- **acute:** supportive (hydration, rest, analgesia); **exclude dengue before NSAIDs**
- **complication: persistent arthralgia or inflammatory arthritis** may need follow-up
- most acute infections improve; **chronic musculoskeletal symptoms are the key distinction**

🔁 **Recap: Chikungunya.**

- **Organism:** alphavirus (Togaviridae), *Aedes aegypti/albopictus*.
- **Clinical: severe polyarthralgia**, sometimes chronic.
- **Testing:** PCR early, serology later.
- **Treatment:** supportive; NSAIDs only after dengue excluded.

🔗 **Link:** the last illness is **not vector-borne**. It is a **zoonosis caught from raw milk and livestock**, which is highly relevant in **Saudi Arabia**. It was already on the FUO list in Part 1.

▶️ **Up next: Brucellosis.**

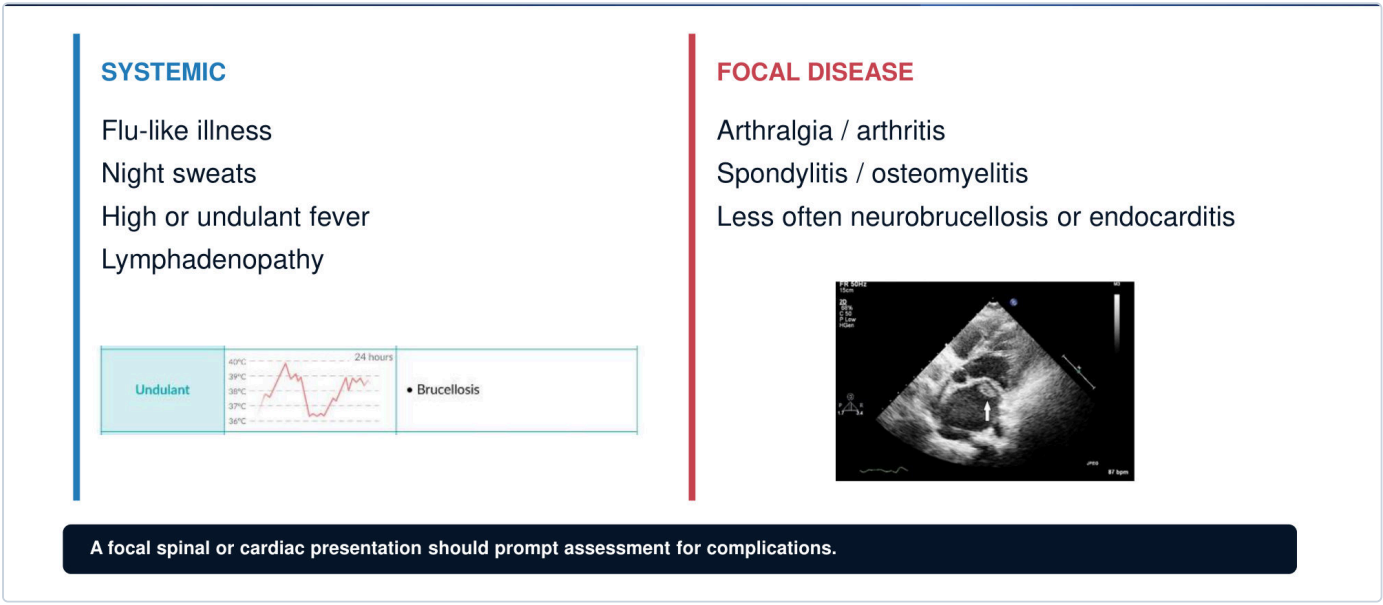
2.14 Brucellosis: Overview (Slide 24) ⭐⭐

Bacterium	Exposures
***Brucella* species**	⭐ **Unpasteurised dairy products**
Small Gram-negative coccobacilli	**Contact with infected animals / animal tissues**
Zoonotic infection	**Occupational exposure** (e.g. **veterinarians**) *[added: farmers, butchers, lab workers]*

⭐ **In Saudi clinical practice, ask specifically about raw milk and livestock exposure.**

- *[added]* Raw **camel** milk is a classic local source.
- *[added] B. melitensis* (goats, sheep, camels) is the most common and most virulent species.

2.15 Brucellosis: Clinical Features & Diagnosis (Slide 25) ⭐



Slide 25: Undulant fever curve; echocardiogram (endocarditis)

Systemic	Focal disease
Flu-like illness	**Arthralgia / arthritis** *[added: sacroiliitis is classic]*
⭐ **Night sweats** *[added: classically malodorous]*	⭐ **Spondylitis / osteomyelitis** *[added: lumbar spine; vs TB thoracic]*
High or undulant fever	Less often **neurobrucellosis** or **endocarditis**
Lymphadenopathy *[added: + hepatosplenomegaly]*	*[added]* Epididymo-orchitis

⭐ A **focal spinal or cardiac presentation** should prompt assessment for **complications**.

2.16 Brucellosis: Diagnosis & Management (Slide 26) ⭐⭐

Diagnosis	Treatment
Serology: screening and confirmatory assays, **including agglutination** *[added: Rose Bengal screen; standard tube agglutination ≥ 1:160]*	⭐ **Uncomplicated adult disease: doxycycline + rifampin, usually ≥ 6 weeks**
Blood cultures when indicated *[added: slow-growing; prolonged incubation]*	Regimens vary with **pregnancy, age, focal complications**

Diagnosis	Treatment
⭐ **Inform the laboratory of suspected *Brucella*!** *[added: a leading cause of lab-acquired infection]*	*[added]* Alternative: doxy + **streptomycin/gentamicin**; spondylitis ≥ 12 weeks; pregnancy → rifampin ± TMP-SMX; children < 8 → TMP-SMX + rifampin

2.17 Approach to Fever (Slides 27–30)

Identical to Part 1, slides 31–34. See the Part 1 notes (§2.18). The same slide errors are repeated:

- **"Sinusitis" twice**
- **"Rheumatoid fever"** for rheumatic fever
- **sarcoidosis** under malignancies
- "**culutres**"

Key history for this lecture:

- **travel** (dengue, Zika, chikungunya)
- **contact with animals** (brucellosis)
- **sexual partner's travel** (Zika)

2.18 To Summarise (Slide 31)



Slide 31: Summary of the four infections

	Key clue	Testing
Dengue	**Bleeding / capillary leak**	**NS1 / NAAT early; IgM later**
Zika	**Conjunctivitis + pregnancy risk**	NAAT / serology per exposure
Chikungunya	**Severe polyarthralgia**	**PCR early; serology later**
Brucellosis	**Raw dairy + night sweats**	**Serology ± cultures**

3. High-Yield Comparison Tables

Table A: The Three Aedes-Borne Arboviruses

[slides + added]

	Dengue	Zika	Chikungunya
Family / genus	***Flavivirus***	***Flavivirus***	**Togaviridae / alphavirus**
Vector	*Ae. aegypti*	*Ae. aegypti* (+ **sexual, transplacental**)	*Ae. aegypti* **+ *Ae. albopictus***
Fever	**High**	**Low-grade**	High, abrupt
Signature	**Retro-orbital pain, leukopenia, thrombocytopenia, bleeding/leak**	⭐ **Non-purulent conjunctivitis**, pruritic rash	⭐ **Severe polyarthralgia** (can be chronic)
Big complication	**Plasma leak → shock** (critical phase)	**Congenital Zika (microcephaly)**, **GBS**	**Chronic arthritis**
Early test	**NS1 or RT-PCR** (+ IgM), days 0–7	NAAT/serology per exposure	**RT-PCR**
Late test	**IgM**	Serology	**Serology**
NSAIDs	❌ **Avoid**	Only after **excluding dengue**	Only after **excluding dengue**
Thrombocytopenia	⭐ **Marked**	Mild/rare *[added]*	Mild *[added]*

Table B: Dengue Classes → Disposition

Class	Features	Disposition (slide 14)
Without warning signs	Fever + ≥ 2 findings; tolerates oral fluids; urine ≥ 6-hourly	**Home**: oral fluids, paracetamol, return precautions
With warning signs	Any warning sign **or** comorbidity **or** social risk	**Observe/admit**, titrated IV isotonic fluids
Severe	Shock/resp distress from leak, severe bleeding, AST/ALT ≥ 1000, CNS, heart	**Admit/ICU**, fluids, **blood products for bleeding**

Table C: Dengue Test Timing

Day of illness	Test
0–7	**RT-PCR (NAAT) or NS1** + IgM
> 7	**IgM**
Any	**Single IgG: not diagnostic**

Table D: Brucellosis vs Spinal TB *[added, bridge to Kotb TB]*

	Brucellar spondylitis	Tuberculous (Pott's)
Exposure	**Raw dairy, livestock**	TB contact, endemic area
Spine level	**Lumbar** (L4–5)	**Thoracic / thoracolumbar**
Bone destruction	Less; disc preserved early	**Marked**, gibbus, paraspinal "cold" abscess
Diagnosis	**Serology (agglutination)**, blood culture	Biopsy, AFB, culture, PCR
Treatment	**Doxy + rifampin** (± aminoglycoside), ≥ 12 weeks	RIPE, prolonged

4. Rules, Exceptions, Patterns

Rule	Exception / nuance
Fever settling = improvement	❌ **Dengue: the critical phase starts at defervescence**
Severe dengue = secondary infection	Usually (**heterologous**), **but can occur in primary infection**
IgG proves infection	❌ **Single IgG not useful** for acute dengue
NSAIDs for fever/arthralgia	❌ Not in dengue; for Zika and chikungunya only **after excluding dengue**
Thrombocytopenia → give platelets	❌ **No prophylactic platelets**; blood products only for **significant bleeding**
Zika = mosquito-borne only	❌ Also **sexual** and **transplacental** → the non-travelling pregnant partner is at risk
Chikungunya = flavivirus like dengue/Zika	❌ It is an **alphavirus (Togaviridae)**
Fever resolves → chikungunya over	❌ **Arthralgia can persist months+**
Brucellosis treatment = 2 weeks	❌ **≥ 6 weeks** of doxy + rifampin
Send blood cultures routinely	⚠️ For *Brucella*, **tell the lab** (biohazard)
Dengue without warning signs → home	Unless **comorbidity** (pregnancy, infancy, elderly, DM, renal failure) or **social** risk → treated as "with warning signs"

Pattern recognition:

- **traveller + fever + retro-orbital pain + leukopenia + platelets ↓ + gum bleeding → dengue (with warning signs)**
- **mild fever + conjunctivitis + pruritic rash ± pregnancy/partner travel → Zika**
- **fever + incapacitating symmetric polyarthralgia → chikungunya**
- **raw milk/livestock + undulant fever + night sweats + back pain → brucellosis**

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5. Clinical Correlations

- **Dengue fluid strategy:**
  - the danger is **capillary leak** during the critical phase
  - give **titrated isotonic fluids** guided by **Hct, vital signs, urine output**
  - *[added]* overload in the recovery phase → pulmonary oedema; reduce fluids once leak stops
  - *[added]* **rising Hct + falling BP = leak**; **falling Hct + instability = bleeding** → transfuse
- **Analgesia in febrile travellers:** use **paracetamol** until dengue is excluded (Zika and chikungunya slides both say this).
- **Pregnancy counselling:**
  - avoid Zika-notice areas
  - **partners returning from endemic areas → condoms/abstinence**
  - *[added]* CDC: men ≥ 3 months, women ≥ 2 months before conception
- **Brucellosis in Saudi Arabia:**
  - endemic
  - take **raw camel/goat milk** and **livestock** history from every patient with prolonged fever or back pain
  - **warn the lab**

- **Neurological: GBS after Zika** → ascending weakness, areflexia *[added: IVIG/plasmapheresis]*.

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6. Rapid Revision (Last-Day)

- **Dengue basics:**
  - **DENV 1–4**, ***Flavivirus***, ***Aedes aegypti***
  - classes: **without warning signs / with warning signs / severe** (WHO 2009)
- **Dengue phases:**
  - **febrile** (days 1–3)
  - ⭐ **critical (at defervescence: leak, shock, ↑Hct)**
  - **recovery** (reabsorption)
- **Dengue without warning signs:**
  - fever + endemic exposure + **≥ 2** of N/V, rash, headache/**retro-orbital** pain, myalgia/arthralgia, **leukopenia**, **positive tourniquet**
  - must tolerate oral fluids and pass urine 6-hourly
- **Warning signs:**
  - abdominal pain, persistent vomiting, **ascites/effusion**, **mucosal bleed**, lethargy/restlessness, **liver > 2 cm**, **↑Hct + ↓plt**
  - or comorbidity (pregnancy, infancy, old age, DM, renal failure) or social risk
- **Severe dengue:**
  - leak → **shock/resp distress**; **severe bleeding**; organ failure (**AST/ALT ≥ 1000**, CNS, heart)
  - ↑ after **heterologous secondary** infection
- **Dengue labs:**
  - leukopenia, thrombocytopenia, ↑AST, Hct trend
  - **days 0–7: RT-PCR or NS1 + IgM**; **> day 7: IgM**; **single IgG useless**
- **Dengue treatment:**
  - **oral fluids + paracetamol + follow-up** → or **admit + titrated IV isotonic fluids, monitor vitals/UO/Hct, blood for bleeding**
  - **no aspirin/NSAIDs, no prophylactic platelets**
- **Zika basics:**
  - **flavivirus** (+ssRNA, enveloped); **Brazil 2015**
  - spread by ***Ae. aegypti*, transplacental, sexual**
- **Zika clinical:**
  - mostly asymptomatic; **low-grade fever, non-purulent conjunctivitis, pruritic maculopapular rash**, arthralgia
  - complications: **congenital Zika (microcephaly, growth restriction, loss)**; **GBS**
- **Zika management:**
  - supportive; **exclude dengue before NSAIDs**
  - **avoid travel in pregnancy; condoms/abstain**; preconception counselling
- **Chikungunya:**
  - **alphavirus (Togaviridae)**; ***Ae. aegypti + Ae. albopictus***; dengue coinfection possible
  - **severe polyarthralgia** that **can persist months**
  - **PCR early, serology later**
  - supportive; NSAIDs after dengue excluded
- **Brucellosis:**

- **small Gram-negative coccobacilli**, zoonosis
- **unpasteurised dairy**, animals, vets; **ask about raw milk/livestock in Saudi**
- flu-like, **night sweats**, **undulant fever**, lymphadenopathy; arthritis, **spondylitis**, neurobrucellosis, **endocarditis**
- **serology (agglutination)** ± blood culture; **inform the lab**
- **doxycycline + rifampin ≥ 6 weeks**

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7. Exam-Style Questions

Q1. (Case Scenario 1) A 26-year-old primigravida at 22 weeks has a stillbirth with microcephaly. Placental RT-PCR shows a flavivirus. She hasn't travelled; her husband, a flight attendant, often visits South America and Southeast Asia. Which measure would most likely have prevented this? A. Avoid unpasteurised dairy · **B. Condoms** · C. cART · D. Avoid undercooked meat · E. Vaccine

> **Answer: B** *[reasoning added; key not on slides]*.
> - **Congenital Zika** via **sexual transmission** from the travelling partner, then transplacental spread.
> - Slide 20: **condoms or abstinence** after possible exposure.
> - A = *Listeria/Brucella*; C = HIV; D = *Toxoplasma*; E = no Zika vaccine exists.

Q2. (Case Scenario 2) A 25-year-old returned from the Philippines 4 days ago. He has 4 days of fever (39.4 °C), arthralgia, myalgia, headache and **retro-orbital pain**; today his **gums bled**. Diffuse maculopapular rash; **WBC 3,200**, **platelets 89,000**; flavivirus detected. Causal pathogen? A. Chikungunya · B. Zika · C. Hepatitis C · **D. Dengue virus**

> **Answer: D** *[reasoning added; key not on slides]*.
> - Retro-orbital pain, leukopenia, thrombocytopenia and **mucosal bleeding** → **dengue**.
> - **Gum bleeding = a warning sign** → **observe/admit**, titrated fluids, monitor Hct, avoid NSAIDs.
> - **Why not the others:**
> - chikungunya is an **alphavirus** (and joint-dominant)
> - Zika is mild with conjunctivitis and rarely causes bleeding
> - HCV is a flavivirus but not an acute febrile haemorrhagic illness

Q3. A dengue patient's fever settles on day 4. He now has abdominal pain, Hct rising from 40 % to 48 %, and platelets 45,000/mm³. BP is normal. Best management? A. Discharge, since the fever has resolved · B. Prophylactic platelet transfusion · **C. Admit, carefully titrated IV isotonic fluids, monitor vitals/UO/Hct** · D. Ibuprofen for pain · E. Start IgG testing

> **Answer: C** (slides 9, 11, 14).
> - **Defervescence = critical phase**: warning signs (abdominal pain, ↑Hct + ↓plt) mean plasma leak.
> - No prophylactic platelets; no NSAIDs; **IgG is not diagnostic**.

Q4. A patient presents on day 3 of fever after returning from Southeast Asia, and you suspect dengue. Best confirmatory test? A. Single dengue IgG · **B. NS1 antigen or RT-PCR (with IgM)** · C. IgM only · D. Tourniquet test alone · E. Blood culture

> **Answer: B** (slide 13).
> - **Days 0–7: NAAT/NS1 + IgM**; after day 7, IgM. **Single IgG is useless.**

Q5. A 40-year-old Saudi shepherd drinks raw camel milk. He has 6 weeks of undulant fever, drenching night sweats, low-back pain and lumbar tenderness. Which is correct? A. Treat with doxycycline for 7 days · **B. Serology (agglutination), warn the lab, doxycycline + rifampin ≥ 6 weeks (longer for spondylitis)** · C. Ceftriaxone monotherapy · D. Start anti-TB therapy without testing · E. NSAIDs and reassurance

> **Answer: B** (slides 24–26).
> - **Brucellosis with spondylitis**. Treatment: **doxy + rifampin ≥ 6 weeks**.
> - *[added]* Focal spinal disease needs **≥ 12 weeks**, often with an aminoglycoside.
> - **Inform the lab.**

Q6. Which arbovirus is an alphavirus, transmitted by both *Ae. aegypti* and *Ae. albopictus*, and causes arthralgia that can persist for months? A. Dengue · B. Zika · **C. Chikungunya** · D. Yellow fever · E. West Nile

> **Answer: C** (slides 21–23).

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8. Exam Pitfalls & Traps

Trap	Correct thinking
Fever gone → dengue improving → discharge	❌ **Critical phase at defervescence**; reassess, check Hct/platelets
Thrombocytopenia → prophylactic platelets	❌ **Not given**; blood products only for significant bleeding
Ibuprofen/aspirin for dengue fever	❌ **Paracetamol only**
Positive dengue IgG = acute dengue	❌ **Single IgG not useful**; use **NS1/PCR ≤ day 7**, **IgM** later
IgM on day 2 is enough	⚠️ Early IgM may be negative; **NS1/PCR** is the early test
Severe dengue only in second infections	❌ More likely after **heterologous secondary infection**, but **can occur in primary**
Zika needs a mosquito bite	❌ **Sexual + transplacental**; the non-travelling partner can be infected
Zika: give NSAIDs for arthralgia	⚠️ **Exclude dengue first**
Chikungunya is a flavivirus	❌ **Alphavirus (Togaviridae)**
Dengue + chikungunya are mutually exclusive	❌ **Coinfection occurs** (shared *Aedes*)
Brucella: routine cultures, short course	❌ **Warn the lab**; **doxy + rifampin ≥ 6 weeks**
Brucella is Gram-positive	❌ **Small Gram-negative coccobacillus**
No travel → no travel illness	❌ **Ask about the partner's travel** (Zika) and **raw dairy/livestock** (brucellosis is local in Saudi)